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Abstract

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I. CANONICAL ANTI COMMUTATION RELATIONSHIP (CAR)

The Canonical Anticommutation Relationships are for a given conb $\{|\varphi_j\rangle | 1 \leq j \leq r\}$ of $\mathcal{H}^1$ as

$$\{a_j, a_k^\dagger\} = \delta_{jk} I_{\mathcal{F}} \quad (1a)$$

$$\{a_j^\dagger, a_k^\dagger\} = \{a_j, a_k\} = 0 \quad (1b)$$

$$1 \leq j \leq r, \quad (1c)$$

where

$$\{A, B\} = AB + BA, \quad (2)$$

the creation operators $\{a_j^\dagger | 1 \leq j \leq r\}$ are defined by their action on *normalized* vectors $|\Psi\rangle \in \mathcal{H}_{\mathcal{F}}$

$$a_j^\dagger |\Psi\rangle = |\varphi_j \wedge \Psi\rangle \quad (3)$$

and the annihilation operators $\{a_j | 1 \leq j \leq r\}$ are defined as their adjoints i.e.

$$a_j = \left(a_j^\dagger\right)^\dagger, \quad 1 \leq j \leq r. \quad (4)$$

1. Comments

- The operators $\{a_j^\dagger | 1 \leq j \leq r\}$ are bounded so have bounded adjoints $\{a_j | 1 \leq j \leq r\}$ and hence

$$a_j^\dagger = (a_j)^\dagger. \quad (5)$$

- In particular it follows from the definition eq. (3) and eq. (1a) that

$$\begin{aligned} a_j^\dagger |\phi\rangle &= |\varphi_j\rangle \\ a_j |\phi\rangle &= 0 \\ 1 \leq j \leq r. \end{aligned} \tag{6}$$

This follows from

$$\langle \phi | a_j^\dagger a_j \phi \rangle = \langle \phi | (I_{\mathcal{F}} - a_j a_j^\dagger) \phi \rangle = 0, \tag{7}$$

which implies that

$$a_j |\phi\rangle = 0. \tag{8}$$

One can set the operators $\{a_j^\dagger | 1 \leq j \leq r\}$ as the values of the map

$$\Phi^\dagger : \mathcal{H}^1 \rightarrow \mathcal{B}(\mathcal{H}_{\mathcal{F}}) \tag{9}$$

i.e.

$$a_j^\dagger = \Phi^\dagger(|\varphi_j\rangle) \tag{10}$$

which leads to a more general (basis free) form of the CAR

$$\{\Phi(|u\rangle), \Phi^\dagger(|v\rangle)\} = \langle u | v \rangle I_{\mathcal{F}} \tag{11a}$$

$$\{\Phi^\dagger(|u\rangle), \Phi^\dagger(|v\rangle)\} = \{\Phi(|u\rangle), \Phi(|v\rangle)\} = 0 \tag{11b}$$

$$\forall |u\rangle, |v\rangle \in \mathcal{H}^1. \tag{11c}$$

The field operator $\Phi^\dagger$ is the restriction of the second quantization map, Γ , to $\mathcal{H}^1$ i.e.

$$\Phi^\dagger = \Gamma|_{\mathcal{H}^1}. \tag{12}$$

II. CLIFFORD ALGEBRA

One can define the self adjoint operators

$$x_j = \frac{1}{\sqrt{2}} (a_j^\dagger + a_j) = x_j^\dagger \tag{13}$$

$$x_{j+r} = \frac{i}{\sqrt{2}} (a_j^\dagger - a_j) = x_{j+r}^\dagger \tag{14}$$

so that

$$a_j^\dagger = \frac{1}{\sqrt{2}} (x_j - ix_{j+r}) \quad (15)$$

$$a_j = \frac{1}{\sqrt{2}} (x_j + ix_{j+r}). \quad (16)$$

$$\frac{1}{\sqrt{2}} (a_j^\dagger + a_j) = \frac{1}{2} ((x_j - ix_{j+r}) + (x_j + ix_{j+r})) = x_j \quad (17)$$

The self adjoint operators are involutions

$$x_j^2 = \frac{1}{\sqrt{2}} (a_j^\dagger + a_j) \frac{1}{\sqrt{2}} (a_j^\dagger + a_j) = \frac{1}{2} I_{\mathcal{F}} \quad (18)$$

$$x_{j+r}^2 = \frac{i}{\sqrt{2}} (a_j^\dagger - a_j) \frac{i}{\sqrt{2}} (a_j^\dagger - a_j) = \frac{1}{2} I_{\mathcal{F}} \quad (19)$$

The products for $1 \leq j \neq k \leq r$ are

$$x_j x_k = \frac{1}{\sqrt{2}} (a_j^\dagger + a_j) \frac{1}{\sqrt{2}} (a_k^\dagger + a_k) = \frac{1}{2} (a_j a_k^\dagger + a_j a_k + a_j^\dagger a_k^\dagger + a_j^\dagger a_k) \quad (20)$$

$$x_k x_j = \frac{1}{\sqrt{2}} (a_k^\dagger + a_k) \frac{1}{\sqrt{2}} (a_j^\dagger + a_j) = \frac{1}{2} (a_k^\dagger a_j + a_k a_j + a_k^\dagger a_j^\dagger + a_k a_j^\dagger) \quad (21)$$

hence

$$x_j x_k = -x_k x_j. \quad (22)$$

The products for $r+1 \leq j \neq k \leq 2r$ are

$$x_{j+r} x_{k+r} = \frac{i}{\sqrt{2}} (a_j^\dagger - a_j) \frac{i}{\sqrt{2}} (a_k^\dagger - a_k) = \frac{1}{2} (-a_j^\dagger a_k^\dagger + a_j^\dagger a_k + a_j a_k^\dagger - a_j a_k) \quad (23)$$

$$x_{k+r} x_{j+r} = \frac{i}{\sqrt{2}} (a_k^\dagger - a_k) \frac{i}{\sqrt{2}} (a_j^\dagger - a_j) = \frac{1}{2} (-a_k^\dagger a_j^\dagger + a_k^\dagger a_j + a_k a_j^\dagger - a_k a_j) \quad (24)$$

hence

$$x_{j+r} x_{k+r} = -x_{k+r} x_{j+r}. \quad (25)$$

The products for $1 \leq j \leq r; r+1 \leq k \leq 2r$

$$x_j x_{k+r} = \frac{1}{\sqrt{2}} (a_j^\dagger + a_j) \frac{i}{\sqrt{2}} (a_k^\dagger - a_k) = \frac{1}{2} (ia_j^\dagger a_k^\dagger - ia_j^\dagger a_k + ia_j a_k^\dagger - ia_j a_k) \quad (26)$$

$$x_{k+r} x_j = \frac{i}{\sqrt{2}} (a_k^\dagger - a_k) \frac{1}{\sqrt{2}} (a_j^\dagger + a_j) = (x_j x_{k+r})^\dagger = \frac{1}{2} (ia_k^\dagger a_j^\dagger - ia_k^\dagger a_j + ia_k a_j^\dagger - ia_k a_j) \quad (27)$$

hence

$$x_j x_{k+r} = -x_{k+r} x_j. \quad (28)$$

The products for $1 \leq j \leq r$ are

$$x_j^2 = x_j x_j = \frac{1}{\sqrt{2}} (a_j^\dagger + a_j) \frac{1}{\sqrt{2}} (a_j^\dagger + a_j) = \frac{1}{2} (a_j a_j^\dagger + a_j a_j + a_j^\dagger a_j^\dagger + a_j^\dagger a_j) = \frac{1}{2} I_{\mathcal{F}} \quad (29)$$

$$x_{j+r}^2 = x_{j+r} x_{j+r} = \frac{i}{\sqrt{2}} (a_j^\dagger - a_j) \frac{i}{\sqrt{2}} (a_j^\dagger - a_j) \quad (30)$$

$$= -\frac{1}{2} (a_j^\dagger a_j^\dagger - a_j^\dagger a_j - a_j a_j^\dagger + a_j a_j) = \frac{1}{2} I_{\mathcal{F}}. \quad (31)$$

Thus the self adjoint operators $\{x_j | 1 \leq j \leq 2r\}$ satisfy the CAR's

$$\{x_j, x_k\} = \delta_{jk} I_{\mathcal{F}}; \quad 1 \leq i, j \leq 2r. \quad (32)$$

The Fermi Fock algebra $\mathcal{F}$, which is representation of a CAR algebra, is generated by complex polynomials of $\{x_j | 1 \leq j \leq 2r\}$ that satisfy the conditions

$$x_j x_k = -x_k x_j, \quad 1 \leq j < k \leq 2r \quad (33)$$

$$x_j^2 = \frac{1}{2} I_{\mathcal{F}}, \quad 1 \leq j \leq 2r \quad (34)$$

or equivalently

$$\{x_j, x_k\} = \delta_{jk} I_{\mathcal{F}}; \quad 1 \leq i, j \leq 2r, \quad (35)$$

thus is a *Clifford Algebra*.

One can introduce a scalar product by considering grade 1 operators C , (these are generators of the algebra), and expressing them as

$$C = \sum_{1 \leq j \leq 2r} c_j x_j \quad (36)$$

and

$$(C|D) = (D|C) = \{C, D\} = \sum_{1 \leq j, k \leq 2r} c_j d_k \{x_j, x_k\} = \sum_{1 \leq j \leq 2r} c_j d_j. \quad (37)$$

The invariance group of the scalar product $(C|D)$ is $SO(2r, \mathbb{C})$. One can define a quadratic function Q by

$$Q(C) = (C|C) \quad (38)$$

and the polarization identity gives

$$(C|D) = \frac{1}{2} (Q(C+D) - Q(C) - Q(D)). \quad (39)$$

A. Canonical Transformations

The transformation

$$T : \mathcal{F} \rightarrow \mathcal{F} \quad (40)$$

given by

$$T \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{a} \end{pmatrix} = \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{a} \end{pmatrix} \mathbb{T} = \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{a} \end{pmatrix} \begin{pmatrix} \mathbb{U} & \bar{\mathbb{V}} \\ \mathbb{V} & \bar{\mathbb{U}} \end{pmatrix} = \begin{pmatrix} \mathbf{b}^\dagger & \mathbf{b} \end{pmatrix} \quad (41)$$

that preserves the CAR's is called a *canonical transformation*. Thus

$$b_k^\dagger = \sum_{1 \leq j \leq r} \left(a_j^\dagger U_{jk} + a_j V_{jk} \right) \quad (42)$$

$$b_k = \sum_{1 \leq j \leq r} \left(a_j^\dagger \bar{V}_{jk} + a_j \bar{U}_{jk} \right) \quad (43)$$

and

$$\delta_{kl} = \left\{ b_k^\dagger, b_l \right\} = \left\{ \sum_{1 \leq j \leq r} \left(a_j^\dagger U_{jk} + a_j V_{jk} \right), \sum_{1 \leq m \leq r} \left(a_m^\dagger \bar{V}_{ml} + a_m \bar{U}_{ml} \right) \right\} \quad (44)$$

$$= \sum_{1 \leq j, m \leq r} \left(\left\{ a_j^\dagger U_{jk}, a_m^\dagger \bar{V}_{ml} \right\} + \left\{ a_j^\dagger U_{jk}, a_m \bar{U}_{ml} \right\} + \left\{ a_j V_{jk}, a_m^\dagger \bar{V}_{ml} \right\} + \left\{ a_j V_{jk}, a_m \bar{U}_{ml} \right\} \right) \quad (45)$$

$$= \sum_{1 \leq j, m \leq r} \left(U_{jk} \left\{ a_j^\dagger, a_m^\dagger \right\} \bar{V}_{ml} + U_{jk} \left\{ a_j^\dagger, a_m \right\} \bar{U}_{ml} + V_{jk} \left\{ a_j, a_m^\dagger \right\} \bar{V}_{ml} + V_{jk} \left\{ a_j, a_m \right\} \bar{U}_{ml} \right) \quad (46)$$

$$= \sum_{1 \leq j, m \leq r} \left(U_{jk} \delta_{jm} \bar{U}_{ml} + V_{jk} \delta_{jm} \bar{V}_{ml} \right) = \sum_{1 \leq j \leq r} \left(U_{jk} \bar{U}_{jl} + V_{jk} \bar{V}_{jl} \right) = \sum_{1 \leq j \leq r} \left(U_{lj}^\dagger U_{jk} + V_{lj}^\dagger V_{jk} \right) \quad (47)$$

$$= (U^\dagger U)_{lk} + (V^\dagger V)_{lk}. \quad (48)$$

In the same way

$$\delta_{lk} = \left\{ b_k, b_l^\dagger \right\} = (U^\dagger U)_{kl} + (V^\dagger V)_{kl}. \quad (49)$$

$$0 = \left\{ b_k^\dagger, b_l^\dagger \right\} = \left\{ \sum_{1 \leq j \leq r} \left(a_j^\dagger U_{jk} + a_j V_{jk} \right), \sum_{1 \leq m \leq r} \left(a_m^\dagger U_{ml} + a_m V_{ml} \right) \right\} \quad (50)$$

$$= \sum_{1 \leq j, m \leq r} \left(U_{jk} \delta_{jm} V_{ml} + V_{jk} \delta_{jm} U_{ml} \right) = \sum_{1 \leq j \leq r} \left(V_{lj}^t U_{jk} + U_{lj}^t V_{jk} \right) \quad (51)$$

$$= (V^t U)_{lk} + (U^t V)_{lk} \quad (52)$$

and

$$0 = \{b_k, b_l\} = (V^\dagger \bar{U})_{lk} + (U^\dagger \bar{V})_{lk}. \quad (53)$$

In matrix notation this gives

$$\left\{ \begin{pmatrix} \mathbf{b} \\ \mathbf{b}^\dagger \end{pmatrix}, \begin{pmatrix} \mathbf{b}^\dagger & \mathbf{b} \end{pmatrix} \right\} = \begin{pmatrix} \{\mathbf{b}, \mathbf{b}^\dagger\} & \{\mathbf{b}, \mathbf{b}\} \\ \{\mathbf{b}^\dagger, \mathbf{b}^\dagger\} & \{\mathbf{b}^\dagger, \mathbf{b}\} \end{pmatrix} = \begin{pmatrix} \mathbb{I}_r & \mathbb{O}_r \\ \mathbb{O}_r & \mathbb{I}_r \end{pmatrix}$$

and

$$\left\{ \begin{pmatrix} \mathbf{b} \\ \mathbf{b}^\dagger \end{pmatrix}, \begin{pmatrix} \mathbf{b}^\dagger & \mathbf{b} \end{pmatrix} \right\} \quad (54)$$

$$= \begin{pmatrix} \mathbf{U} & \bar{\mathbf{V}} \\ \mathbf{V} & \bar{\mathbf{U}} \end{pmatrix}^\dagger \left\{ \begin{pmatrix} \mathbf{a} \\ \mathbf{a}^\dagger \end{pmatrix}, \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{a} \end{pmatrix} \right\} \begin{pmatrix} \mathbf{U} & \bar{\mathbf{V}} \\ \mathbf{V} & \bar{\mathbf{U}} \end{pmatrix} \quad (55)$$

$$= \begin{pmatrix} \mathbf{U} & \bar{\mathbf{V}} \\ \mathbf{V} & \bar{\mathbf{U}} \end{pmatrix}^\dagger \begin{pmatrix} \{\mathbf{a}, \mathbf{a}^\dagger\} & \{\mathbf{a}, \mathbf{a}\} \\ \{\mathbf{a}^\dagger, \mathbf{a}^\dagger\} & \{\mathbf{a}^\dagger, \mathbf{a}\} \end{pmatrix} \begin{pmatrix} \mathbf{U} & \bar{\mathbf{V}} \\ \mathbf{V} & \bar{\mathbf{U}} \end{pmatrix} \quad (56)$$

$$= \begin{pmatrix} \mathbf{U} & \bar{\mathbf{V}} \\ \mathbf{V} & \bar{\mathbf{U}} \end{pmatrix}^\dagger \begin{pmatrix} \mathbb{I}_r & \mathbb{O}_r \\ \mathbb{O}_r & \mathbb{I}_r \end{pmatrix} \begin{pmatrix} \mathbf{U} & \bar{\mathbf{V}} \\ \mathbf{V} & \bar{\mathbf{U}} \end{pmatrix} \quad (57)$$

$$= \begin{pmatrix} \mathbf{U} & \bar{\mathbf{V}} \\ \mathbf{V} & \bar{\mathbf{U}} \end{pmatrix}^\dagger \begin{pmatrix} \mathbf{U} & \bar{\mathbf{V}} \\ \mathbf{V} & \bar{\mathbf{U}} \end{pmatrix} = \begin{pmatrix} \mathbf{U}^\dagger & \mathbf{V}^\dagger \\ \bar{\mathbf{V}}^\dagger & \bar{\mathbf{U}}^\dagger \end{pmatrix} \begin{pmatrix} \mathbf{U} & \bar{\mathbf{V}} \\ \mathbf{V} & \bar{\mathbf{U}} \end{pmatrix} = \begin{pmatrix} \mathbb{I}_r & \mathbb{O}_r \\ \mathbb{O}_r & \mathbb{I}_r \end{pmatrix}. \quad (58)$$

Hence $\begin{pmatrix} \mathbf{U} & \bar{\mathbf{V}} \\ \mathbf{V} & \bar{\mathbf{U}} \end{pmatrix}$ is a unitary matrix and

$$\mathbf{U}^\dagger \mathbf{U} + \mathbf{V}^\dagger \mathbf{V} = \mathbb{I}_r \quad (59)$$

$$\mathbf{U}^\dagger \bar{\mathbf{V}} + \mathbf{V}^\dagger \bar{\mathbf{U}} = \mathbb{O}_r. \quad (60)$$

B. Real Basis of the CAR Fock Algebra

The transformation between the $\{a_j, a_j^\dagger | 1 \leq j \leq r\}$ and $\{x_j | 1 \leq j \leq 2r\}$ bases is given by

$$\begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{a} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \mathbb{I}_r & \frac{i}{\sqrt{2}} \mathbb{I}_r \\ \frac{1}{\sqrt{2}} \mathbb{I}_r & -\frac{i}{\sqrt{2}} \mathbb{I}_r \end{pmatrix}, \quad (61)$$

where

$$\begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix} \in U(2r, \mathbb{R}) \quad (62)$$

i.e.

$$\begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix}^\dagger \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix} = \quad (63)$$

$$\begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{1}{\sqrt{2}}\mathbb{I}_r \\ \frac{-i}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix} = \begin{pmatrix} \mathbb{I}_r & \mathbb{O}_r \\ \mathbb{O}_r & \mathbb{I}_r \end{pmatrix}. \quad (64)$$

The real basis produced by the canonically transformed $\{a_j^\dagger, a_j \mid 1 \leq j \leq r\}$ is

$$\begin{pmatrix} \mathbf{y}_1 & \mathbf{y}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{b}^\dagger & \mathbf{b} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix} \quad (65)$$

$$= \begin{pmatrix} \mathbf{a}^\dagger & \mathbf{a} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix}^\dagger \begin{pmatrix} \mathbb{U} & \bar{\mathbb{V}} \\ \mathbb{V} & \bar{\mathbb{U}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix} \quad (66)$$

$$= \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix}^\dagger \begin{pmatrix} \mathbb{U} & \bar{\mathbb{V}} \\ \mathbb{V} & \bar{\mathbb{U}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix} \quad (67)$$

$$= \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{1}{\sqrt{2}}\mathbb{I}_r \\ \frac{-i}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix} \begin{pmatrix} \mathbb{U} & \bar{\mathbb{V}} \\ \mathbb{V} & \bar{\mathbb{U}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix} \quad (68)$$

$$= \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{U} + \frac{1}{\sqrt{2}}\mathbb{V} & \frac{1}{\sqrt{2}}\bar{\mathbb{V}} + \frac{1}{\sqrt{2}}\bar{\mathbb{U}} \\ \frac{-i}{\sqrt{2}}\mathbb{U} + \frac{i}{\sqrt{2}}\mathbb{V} & \frac{-i}{\sqrt{2}}\bar{\mathbb{V}} + \frac{i}{\sqrt{2}}\bar{\mathbb{U}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix} \quad (69)$$

$$= \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{pmatrix} \begin{pmatrix} \frac{1}{2}\mathbb{U} + \frac{1}{2}\mathbb{V} + \frac{1}{2}\bar{\mathbb{V}} + \frac{1}{2}\bar{\mathbb{U}} & \frac{i}{2}\mathbb{U} - \frac{i}{2}\mathbb{V} + \frac{i}{2}\bar{\mathbb{V}} - \frac{i}{2}\bar{\mathbb{U}} \\ \frac{-i}{2}\mathbb{U} + \frac{i}{2}\mathbb{V} - \frac{i}{2}\bar{\mathbb{V}} + \frac{i}{2}\bar{\mathbb{U}} & \frac{1}{2}\mathbb{U} + \frac{1}{2}\mathbb{V} - \frac{1}{2}\bar{\mathbb{V}} + \frac{1}{2}\bar{\mathbb{U}} \end{pmatrix} \quad (70)$$

$$= \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{pmatrix} \begin{pmatrix} \text{Re}(\mathbb{U} + \mathbb{V}) & \text{Im}(\mathbb{V} + \mathbb{U}) \\ \text{Im}(\mathbb{V} - \mathbb{U}) & \text{Re}(\mathbb{U} - \mathbb{V}) \end{pmatrix}. \quad (71)$$

i.e.

$$\begin{pmatrix} \mathbf{y}_1 & \mathbf{y}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{x}_1 & \mathbf{x}_2 \end{pmatrix} \begin{pmatrix} \text{Re}(\mathbb{U} + \mathbb{V}) & \text{Im}(\mathbb{V} + \mathbb{U}) \\ \text{Im}(\mathbb{V} - \mathbb{U}) & \text{Re}(\mathbb{U} - \mathbb{V}) \end{pmatrix} \quad (72)$$

As both $\begin{pmatrix} \mathbb{U} & \bar{\mathbb{V}} \\ \mathbb{V} & \bar{\mathbb{U}} \end{pmatrix}$ and $\begin{pmatrix} \frac{1}{\sqrt{2}}\mathbb{I}_r & \frac{i}{\sqrt{2}}\mathbb{I}_r \\ \frac{1}{\sqrt{2}}\mathbb{I}_r & -\frac{i}{\sqrt{2}}\mathbb{I}_r \end{pmatrix}$ are unitary one has that

$$\begin{pmatrix} \operatorname{Re}(\mathbb{U} + \mathbb{V}) & \operatorname{Im}(\mathbb{V} + \mathbb{U}) \\ \operatorname{Im}(\mathbb{V} - \mathbb{U}) & \operatorname{Re}(\mathbb{U} - \mathbb{V}) \end{pmatrix} \begin{pmatrix} \operatorname{Re}(\mathbb{U} + \mathbb{V}) & \operatorname{Im}(\mathbb{V} + \mathbb{U}) \\ \operatorname{Im}(\mathbb{V} - \mathbb{U}) & \operatorname{Re}(\mathbb{U} - \mathbb{V}) \end{pmatrix}^\dagger \quad (73)$$

$$= \begin{pmatrix} \operatorname{Re}(\mathbb{U} + \mathbb{V}) & \operatorname{Im}(\mathbb{V} + \mathbb{U}) \\ \operatorname{Im}(\mathbb{V} - \mathbb{U}) & \operatorname{Re}(\mathbb{U} - \mathbb{V}) \end{pmatrix}^\dagger \begin{pmatrix} \operatorname{Re}(\mathbb{U} + \mathbb{V}) & \operatorname{Im}(\mathbb{V} + \mathbb{U}) \\ \operatorname{Im}(\mathbb{V} - \mathbb{U}) & \operatorname{Re}(\mathbb{U} - \mathbb{V}) \end{pmatrix} \quad (74)$$

$$= \mathbb{I}_{2r} \quad (75)$$

and as $\begin{pmatrix} \operatorname{Re}(\mathbb{U} + \mathbb{V}) & \operatorname{Im}(\mathbb{V} + \mathbb{U}) \\ \operatorname{Im}(\mathbb{V} - \mathbb{U}) & \operatorname{Re}(\mathbb{U} - \mathbb{V}) \end{pmatrix}$ is a real matrix

$$\begin{pmatrix} \operatorname{Re}(\mathbb{U} + \mathbb{V}) & \operatorname{Im}(\mathbb{V} + \mathbb{U}) \\ \operatorname{Im}(\mathbb{V} - \mathbb{U}) & \operatorname{Re}(\mathbb{U} - \mathbb{V}) \end{pmatrix}^\dagger = \begin{pmatrix} \operatorname{Re}(\mathbb{U} + \mathbb{V}) & \operatorname{Im}(\mathbb{V} + \mathbb{U}) \\ \operatorname{Im}(\mathbb{V} - \mathbb{U}) & \operatorname{Re}(\mathbb{U} - \mathbb{V}) \end{pmatrix}^t \quad (76)$$

and

$$\begin{pmatrix} \operatorname{Re}(\mathbb{U} + \mathbb{V}) & \operatorname{Im}(\mathbb{V} + \mathbb{U}) \\ \operatorname{Im}(\mathbb{V} - \mathbb{U}) & \operatorname{Re}(\mathbb{U} - \mathbb{V}) \end{pmatrix} \in O(2r, \mathbb{R}) \quad (77)$$

i.e. *the group of canonical transformations is $O(2r, \mathbb{R})$.*

In general

$$X(|u\rangle) = \frac{1}{\sqrt{2}} (\Phi^\dagger(|u\rangle) + \Phi(|u\rangle)) \quad (78)$$

$$Y(|u\rangle) = \frac{i}{\sqrt{2}} (\Phi^\dagger(|u\rangle) - \Phi(|u\rangle)) \quad (79)$$

$$\Phi^\dagger(|u\rangle) = \frac{1}{\sqrt{2}} (X(|u\rangle) - iY(|u\rangle)) \quad (80)$$

$$\Phi(|u\rangle) = \frac{1}{\sqrt{2}} (X(|u\rangle) + iY(|u\rangle)). \quad (81)$$

Hence

$$\{X(|u\rangle), X(|v\rangle)\} = \frac{1}{2} \{(\Phi^\dagger(|u\rangle) + \Phi(|u\rangle)), (\Phi^\dagger(|v\rangle) + \Phi(|v\rangle))\} \quad (82)$$

$$= \{\Phi^\dagger(|u\rangle), \Phi^\dagger(|v\rangle)\} + \{\Phi^\dagger(|u\rangle), \Phi(|v\rangle)\} + \{\Phi(|u\rangle), \Phi^\dagger(|v\rangle)\} + \{\Phi(|u\rangle), \Phi(|v\rangle)\} \quad (83)$$

$$= 0 + \overline{\langle u|v\rangle} I_{\mathcal{F}} + \langle u|v\rangle I_{\mathcal{F}} + 0 = \text{Re} \langle u|v\rangle I_{\mathcal{F}} \quad (84)$$

$$\{X(|u\rangle), Y(|v\rangle)\} = \{(\Phi^\dagger(|u\rangle) + \Phi(|u\rangle)), i(\Phi^\dagger(|v\rangle) - \Phi(|v\rangle))\} \quad (85)$$

$$= i\{\Phi^\dagger(|u\rangle), \Phi^\dagger(|v\rangle)\} - i\{\Phi^\dagger(|u\rangle), \Phi(|v\rangle)\} + i\{\Phi(|u\rangle), \Phi^\dagger(|v\rangle)\} - i\{\Phi(|u\rangle), \Phi(|v\rangle)\} \quad (86)$$

$$= 0 - i\overline{\langle u|v\rangle} I_{\mathcal{F}} + i\langle u|v\rangle I_{\mathcal{F}} = \text{Im} \langle u|v\rangle I_{\mathcal{F}} \quad (87)$$

$$\{Y(|u\rangle), Y(|v\rangle)\} = \{i(\Phi^\dagger(|u\rangle) - \Phi(|u\rangle)), i(\Phi^\dagger(|v\rangle) - \Phi(|v\rangle))\} \quad (88)$$

$$= -\{\Phi^\dagger(|u\rangle), \Phi^\dagger(|v\rangle)\} + \{\Phi^\dagger(|u\rangle), \Phi(|v\rangle)\} + \{\Phi(|u\rangle), \Phi^\dagger(|v\rangle)\} - \{\Phi(|u\rangle), \Phi(|v\rangle)\} \quad (89)$$

$$= 0 + \overline{\langle u|v\rangle} I_{\mathcal{F}} + \langle u|v\rangle I_{\mathcal{F}} + 0 = \text{Re} \langle u|v\rangle I_{\mathcal{F}} \quad (90)$$

and

$$\{X(|u\rangle), X(|u\rangle)\} = \langle u|u\rangle I_{\mathcal{F}} \quad (91)$$

$$\{X(|u\rangle), Y(|u\rangle)\} = 0 \quad (92)$$

$$\{Y(|u\rangle), Y(|v\rangle)\} = \langle u|u\rangle I_{\mathcal{F}} \quad (93)$$

$$\{X(|u_i\rangle), X(|u_j\rangle)\} = \delta_{ij} I_{\mathcal{F}} \quad (94)$$

$$\{X(|u_i\rangle), Y(|u_j\rangle)\} = 0 \quad (95)$$

$$\{Y(|u_i\rangle), Y(|u_j\rangle)\} = \delta_{ij} I_{\mathcal{F}} \quad (96)$$

III. GEOMETRIC ALGEBRA

$$A = \sum_{1 \leq j \leq 2r} \alpha_j x_j; \quad B = \sum_{1 \leq j \leq 2r} \beta_j x_j \quad (97)$$

$$AB = \sum_{1 \leq j, k \leq 2r} \alpha_j \beta_k x_j x_k = \sum_{1 \leq j \leq 2r} \alpha_j \beta_j x_j^2 + \sum_{1 \leq j \neq k \leq 2r} \alpha_j \beta_k x_j x_k = \sum_{1 \leq j \leq 2r} \alpha_j \beta_j + 2 \sum_{1 \leq j < k \leq 2r} \alpha_j \beta_k x_j x_k \quad (98)$$

$$= (A|B) I_{\mathcal{F}} + A \wedge B \quad (99)$$

$$A = \sum_{1 \leq j \leq 2r} \alpha_j x_j = \sum_{1 \leq j \leq r} \{\alpha_j x_j + \alpha_{j+r} x_{j+r}\} = \sum_{1 \leq j \leq r} \left\{ \alpha_j (a_j^\dagger + a_j) + \alpha_{j+r} i (a_j^\dagger - a_j) \right\} \quad (100)$$

$$= \sum_{1 \leq j \leq r} \left\{ \alpha_j a_j^\dagger + \alpha_j a_j + i \alpha_{j+r} a_j^\dagger - i \alpha_{j+r} a_j \right\} = \sum_{1 \leq j \leq r} \left\{ (\alpha_j + i \alpha_{j+r}) a_j^\dagger + (\alpha_j - i \alpha_{j+r}) a_j \right\} \quad (101)$$